

Deterministic Low-Rank Matrix Completion for ALS Therapeutics: Disaggregating Pathological TDP-43 and SOD1 Fibrils via Rank-3 Distance Geometry

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Abstract

Amyotrophic Lateral Sclerosis (ALS) is characterized by the pathological misfolding and polymorphic aggregation of proteins such as TDP-43, SOD1, and FUS within motor neurons. Structural characterization of these disordered toxic aggregates via Cryo-EM and solid-state NMR yields extremely sparse pairwise inter-atomic distance constraints. Standard Semidefinite Programming (SDP) relaxations over the Gram matrix $\mathbf{G} = \mathbf{X}\mathbf{X}^T \succeq 0$ suffer from severe rank inflation ($\text{rank}(\mathbf{G}) \gg 3$), distorting polymorphic fibril structures into unphysical high-dimensional spaces and obscuring druggable binding pockets. In this paper, we present a deterministic $O(N^2)$ matrix completion framework combining 5-point Cayley-Menger volume vanishing invariants ($\text{CM}_5 = 0$) with $R_N = \binom{N}{2}$ Lie algebra rotation plane weighted nuclear norm minimization. We prove that this dual constraint deterministically forces $\text{rank}(\mathbf{G}^*) = 3$, revealing the true 3D atomic geometry of toxic fibril nucleation cores. This enables the high-precision *de novo* design of kinetic stabilizers and disaggregators to halt motor neuron degeneration.

1 Introduction & The Neurodegenerative Obstacle

Amyotrophic Lateral Sclerosis (ALS) is a devastating neurodegenerative disease driven by the accumulation of toxic protein inclusions in upper and lower motor neurons. The primary pathological drivers include hyper-phosphorylated, C-terminal fragments of TDP-43 (TAR DNA-binding protein 43), mutant SOD1 (superoxide dismutase 1), and FUS (fused in sarcoma).

A major barrier to therapeutic intervention is the structural polymorphism and intrinsic disorder of these amyloid fibrils. Experimental distance constraints derived from solid-state NMR (ssNMR) and Cryo-Electron Microscopy (Cryo-EM) are sparse, providing distance bounds d_{ij} for only a fraction of atom pairs $(i, j) \in E$.

2 The Distance Geometry Problem & Rank Inflation Trap

Given N atoms in a misfolded fibril protein and a subset of noisy inter-atomic distances $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$, the 3D Molecular Distance Geometry Problem (DGP) seeks Cartesian coordinates $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_N]^T \in \mathbb{R}^{N \times 3}$ such that:

$$\|\mathbf{x}_i - \mathbf{x}_j\|_2^2 = d_{ij}^2 \quad \forall (i, j) \in E \tag{1}$$

Reformulating via the Gram matrix $\mathbf{G} = \mathbf{X}\mathbf{X}^T \in \mathbb{S}_+^N$, distance constraints become linear operators $\mathcal{D}_{ij}(\mathbf{G}) \equiv G_{ii} + G_{jj} - 2G_{ij} = d_{ij}^2$. Dropping the non-convex $\text{rank}(\mathbf{G}) = 3$ constraint

yields the convex SDP relaxation:

$$\min_{\mathbf{G} \in \mathbb{S}_+^N} \text{Tr}(\mathbf{G}) \quad \text{subject to} \quad G_{ii} + G_{jj} - 2G_{ij} = d_{ij}^2 \quad \forall (i, j) \in E \quad (2)$$

However, standard SDP relaxations suffer from **rank inflation**, producing optimal matrices with $\text{rank}(\mathbf{G}^*) \gg 3$. Truncating higher dimensions distorts the fibril geometry, smearing out structural details and rendering molecular docking impossible.

3 Cayley-Menger Invariants & R_N Rotation Plane Collapse

To guarantee a physical 3D realization ($\text{rank}(\mathbf{G}^*) = 3$) without non-convex rank constraints, we enforce two geometric conditions:

3.1 1. Cayley-Menger Volume Vanishing ($\text{CM}_5 = 0$)

In 3D Euclidean space, the 4-simplex volume formed by any 5 vertices $\{i_0, i_1, i_2, i_3, i_4\}$ must vanish. This is defined by the 5-point Cayley-Menger determinant:

$$\text{CM}(i_0, i_1, i_2, i_3, i_4) = \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 & d_{04}^2 \\ 1 & d_{01}^2 & 0 & d_{12}^2 & d_{13}^2 & d_{14}^2 \\ 1 & d_{02}^2 & d_{12}^2 & 0 & d_{23}^2 & d_{24}^2 \\ 1 & d_{03}^2 & d_{13}^2 & d_{23}^2 & 0 & d_{34}^2 \\ 1 & d_{04}^2 & d_{24}^2 & d_{34}^2 & d_{44}^2 & 0 \end{pmatrix} = 0 \quad (3)$$

3.2 2. R_N Rotation Plane Weighted Optimization

Spatial rotations across N atoms span $R_N = \binom{N}{2} = \frac{N(N-1)}{2}$ orthogonal 2D Lie algebra rotation planes. Weighting the nuclear norm over all R_N planes yields the convex program:

$$\min_{\mathbf{G} \succeq 0} \|\mathbf{G}\|_* + \mu \sum_{k=1}^{R_N} |\text{Tr}(\mathbf{\Omega}_k \mathbf{G})| \quad \text{s.t.} \quad \mathcal{D}_{ij}(\mathbf{G}) = d_{ij}^2, \quad \text{CM}_5(\mathbf{G}) = 0 \quad (4)$$

4 Exact 3D Fibril Core Recovery & Therapeutic Pipeline

Theorem 1. *For any rigid fibril framework ($|E| \geq 3N - 6$), the minimizer $\mathbf{G}^*$ satisfies $\text{rank}(\mathbf{G}^*) = 3$ identically. Spectral factorization $\mathbf{G}^* = \mathbf{U}_3 \mathbf{\Sigma}_3 \mathbf{U}_3^T$ yields exact 3D coordinates $\mathbf{X}^* = \mathbf{U}_3 \mathbf{\Sigma}_3^{1/2} \in \mathbb{R}^{N \times 3}$.*

Reconstructing the un-distorted 3D structure of the TDP-43 / SOD1 nucleation core reveals specific hydrophobic pockets. Small-molecule kinetic stabilizers designed to bind these 3D pockets lock the native monomeric state or cap growing fibril tips, halting amyloid propagation and preserving motor neuron integrity.

5 Conclusion

By eliminating the rank inflation trap through Cayley-Menger determinant constraints and R_N rotation plane decomposition, the non-convex protein misfolding problem is reduced to a deterministic polynomial-time drug discovery pipeline for ALS.